

Your name here
Collaborators:
MATH 300 HW5

A Gentle Reminder: ***feel free to delete this reminder on your own homework*** Please do not use any form of AI while completing your homework – this will only inhibit your ability to think mathematically and write strong proofs. If you have questions, ask your classmates, me, or Varun!

Exercise 1. How many 10-letter words use exactly four A's, three B's, two C's and one D?

Solution.

□

Exercise 2. Let $S = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x \neq 2\}$ and define $f: S \rightarrow S$ by

$$f(x, y) = \left(\frac{y+2}{x-2}, \frac{1}{x-2} \right).$$

Prove that f is injective. Is f surjective? Prove your claim.

Proof.

□

Exercise 3. Prove the following identity via a combinatorial argument (i.e., show that the two equations count the same object):

$$1n + 2(n-1) + 3(n-2) + 4(n-3) + \cdots + (n-1)2 + n1 = \binom{n+2}{3}.$$

Proof.

□

Exercise 4. Give a combinatorial proof of this identity. Hint: What if some number of a group of people wanted to go to an escape room, and among those going, one needed to be the team captain?

$$1\binom{n}{1} + 2\binom{n}{2} + 3\binom{n}{3} + \cdots + (n-1)\binom{n}{n-1} + n\binom{n}{n} = n2^{n-1}$$

Proof.

□

Exercise 5. Use the formula for $\binom{n}{k}$ to give an algebraic proof of the following identity:

$$k\binom{n}{k} = n\binom{n-1}{k-1}.$$

Proof.

□